Functional Requirements Document (FRD)

Project: TeachHub Lagos – A Centralised Digital Resource Platform for Educators

Document Information	Details
Project Name	TeachHub Lagos
Document Type	Functional Requirements Document
Project Domain	Education Technology (EdTech)
My Role	Business Analyst
Project Team	3 members
Primary Users	Teachers and Administrators
Project Status	Functional Proof of Concept

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1. Introduction

1.1 Purpose of This Document

This Functional Requirements Document defines the expected system behaviour and functionality of TeachHub Lagos.

It translates the business needs identified during requirements gathering into specific functional requirements that guide system development, testing, and validation.

The document is intended to provide a shared understanding of the expected system functionality among the Business Analyst, frontend developer, and backend developer.

1.2 Project Overview

TeachHub Lagos is a centralised digital platform designed to help educators create, manage, organise, and share educational resources.

The system provides functionality for user management, lesson management, and educational resource management.

1.3 System Users

The system has two primary user roles:

Teacher

- Update profile information.
- Change account password.
- Create and manage lessons.
- Attach educational resources to lessons.
- Create quizzes and flashcards.
- Preview and publish lesson content.
- Upload, edit, view, and share educational resources.

Administrator

- Add new users.
- Modify and delete user accounts.
- Manage user roles and permissions.
- View content created on the platform.
- Control resource deletion.

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2. Functional Requirements

Functional requirements describe what the system must do to support its intended users.

The requirements are grouped into three core modules.

2.1 Module One: User Control

This module manages user accounts, authentication-related access, roles, and permissions.

Requirement ID	Functional Requirement	User Role	Priority
FR-001	The system shall allow users to register and authenticate.	User	High
FR-002	The system shall allow administrators to add new users.	Administrator	High
FR-003	The system shall allow administrators to modify user accounts.	Administrator	High
FR-004	The system shall allow administrators to delete user accounts.	Administrator	High
FR-005	The system shall allow administrators to modify user roles and permissions.	Administrator	High
FR-006	The system shall allow administrators to view content created on the platform.	Administrator	High
FR-007	The system shall allow teachers to update their profile information.	Teacher	Medium
FR-008	The system shall allow teachers to change their passwords securely.	Teacher	High

Acceptance Criteria

- An authorised administrator can add a new user.
- An administrator can modify an existing user's details.
- An administrator can manage user roles and permissions.
- An administrator can delete a user account.

- A teacher can update their profile information.
- A teacher can change their password through the appropriate account functionality.
- Users can authenticate to access the platform.

2.2 Module Two: Lesson Management

This module enables teachers to create, organise, preview, and publish educational lessons.

Requirement ID	Functional Requirement	User Role	Priority
FR-009	The system shall allow teachers to create lessons.	Teacher	High
FR-010	The system shall allow teachers to enter a lesson title.	Teacher	High
FR-011	The system shall allow teachers to enter lesson plans and objectives.	Teacher	High
FR-012	The system shall allow teachers to specify a subject and grade level for a lesson.	Teacher	High
FR-013	The system shall support rich-text lesson content.	Teacher	High
FR-014	The system shall allow teachers to attach images, videos, and other learning resources to lessons.	Teacher	High
FR-015	The system shall allow teachers to create quizzes containing multiple-choice questions.	Teacher	Medium
FR-016	The system shall allow teachers to create quizzes containing short-answer questions.	Teacher	Medium
FR-017	The system shall allow	Teacher	Medium

	teachers to create flashcards containing images and text.		
FR-018	The system shall allow teachers to preview lesson content before publishing.	Teacher	High
FR-019	The system shall allow teachers to publish lesson content for platform-wide sharing.	Teacher	High
FR-020	The system shall allow teachers to manage their lesson content.	Teacher	High

Acceptance Criteria

- A teacher can create a lesson with a title, plan, subject, grade level, objectives, and content.
- A teacher can attach supported learning resources to a lesson.
- A teacher can create quizzes using the supported question types.
- A teacher can create flashcards containing images and text.
- A teacher can preview lesson content before publishing.
- A teacher can publish lesson content for sharing.

2.3 Module Three: Resource Management

This module provides functionality for uploading, categorising, viewing, and managing educational resources.

Requirement ID	Functional Requirement	User Role	Priority
FR-021	The system shall allow authorised users to upload educational resources.	Permitted User	High
FR-022	The system shall support educational resources such as images, videos, and worksheets.	Permitted User	High
FR-023	The system shall allow resources to be categorised by subject.	Permitted User	High
FR-024	The system shall allow resources to be categorised by grade level.	Permitted User	High
FR-025	The system shall allow authorised users to edit uploaded resources.	Permitted User	Medium
FR-026	The system shall allow authorised users to view uploaded resources.	Permitted User	High
FR-027	The system shall restrict resource deletion to administrators.	Administrator	High

Acceptance Criteria

- An authorised user can upload an educational resource.
- A resource can be categorised by subject and grade level.
- An authorised user can view an uploaded resource.
- An authorised user can edit an uploaded resource.
- Only an administrator can delete a resource.

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3. Non-Functional Requirements

Non-functional requirements describe the quality attributes and operating expectations of the platform.

Requirement ID	Category	Requirement
NFR-001	Usability	The system should provide an intuitive interface that requires minimal technical training.
NFR-002	Navigation	The system should maintain consistent navigation across its interfaces.
NFR-003	Responsiveness	The platform should use a mobile-first design approach.
NFR-004	Security	User accounts and protected functionality should be appropriately secured.
NFR-005	Access Control	Access to administrative functionality should be restricted according to user roles and permissions.
NFR-006	Browser Compatibility	The system should be accessible through modern web browsers.
NFR-007	Performance	The system should provide a usable experience for users who may experience slow internet connectivity.

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4. User Roles and Permissions Matrix

The following matrix summarises the expected access rights based on the project requirements.

System Functionality	Teacher	Administrator
Register and authenticate	Yes	Yes
Update teacher profile	Yes	Not specified
Change teacher password	Yes	Not specified
Add users	No	Yes
Modify user accounts	No	Yes
Delete user accounts	No	Yes
Manage roles and permissions	No	Yes
View platform content	Own/accessible content	All platform content
Create and manage lessons	Yes	Not specified
Create quizzes	Yes	Not specified
Create flashcards	Yes	Not specified
Upload educational resources	Yes	Yes, subject to permissions
Edit and view resources	Yes, subject to permissions	Yes
Delete resources	No	Yes

The matrix reflects the documented responsibilities and permissions. Where the original report does not explicitly define an access rule, it is marked as not specified rather than assumed.

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5. Business Rules

The following business rules support the functional requirements:

Rule ID	Business Rule
BR-001	Users must authenticate to access protected platform functionality.
BR-002	Administrators manage user accounts, roles, and permissions.
BR-003	Teachers are responsible for creating and managing instructional content.
BR-004	Educational resources should be categorised by subject and grade level.
BR-005	Resource deletion is restricted to administrators.
BR-006	Teachers should be able to preview lessons before publishing.
BR-007	Teachers can set created lesson content to private or global mode.

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6. Assumptions and Dependencies

The following considerations were relevant to the system requirements:

- Users have access to a supported web browser.
- Users have a device capable of accessing the web application.
- Internet access is required for normal online platform usage.
- User access is managed through defined roles and permissions.
- The platform relies on its database and application infrastructure to store and manage content.

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7. Requirements Traceability Matrix

This matrix connects business needs to functional requirements and the relevant testing activities.

Business Requirement	Related Functional Requirements	Suggested Test Area
BR-01: Centralised educational resources	FR-021 to FR-026	Resource management
BR-02: Lesson creation and management	FR-009 to FR-020	Lesson management
BR-03: Resource organisation	FR-023 to FR-024	Resource categorisation
BR-04: Secure user authentication	FR-001, FR-008	Authentication
BR-05: Role-based access	FR-002 to FR-006, FR-027	Access control
BR-06: Mobile-responsive experience	NFR-003	Responsive interface
BR-07: Administrative management	FR-002 to FR-006	User administration

The test areas are suggested validation activities derived from the requirements.

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8. Requirements Validation

Requirements validation helps confirm that the specified functionality reflects the intended business needs.

The project report describes module testing, integration testing, and User Acceptance Testing (UAT).

The following validation questions can be used during requirements review:

1. Does the system support the main activities expected of teachers?
2. Can administrators manage users and permissions?
3. Can teachers create and manage lesson content?
4. Can educational resources be uploaded and categorised?
5. Are resource deletion permissions appropriately restricted?
6. Can teachers preview lesson content before publishing?
7. Does the interface reflect the mobile-first usability expectation?

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9. Conclusion

The functional requirements documented in this FRD provide a structured description of the expected behaviour of TeachHub Lagos.

By translating business needs into specific system requirements, the document provides a reference for developers during implementation and for testers during verification and validation.

As the Business Analyst, my contribution involved analysing the user needs, documenting system functionality, clarifying requirements, and supporting the validation of the developed solution.

Document: Functional Requirements Document
Project: TeachHub Lagos
Prepared for: Business Analysis Portfolio